(3)

(Please write your Enrollment Number) Enrollment No. 194

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OFFLINE)
Course Name:<B.Tech (CSE-AI)>, Semester:<>
(May, 2026)

Subject Code: BAI 304	Subject: Deep Learning – II
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
a)	Distinguish between bias and variance in deep learning.	(2.5)
b)	What is batch normalization? State one benefit of using it in deep neural networks.	(2.5)
c)	State any two differences between Batch Gradient Descent and Mini-Batch Gradient Descent.	(2.5)
d)	Name any two optimization methods among Adagrad, RMSProp, Adam, and NAG and state one key idea behind each.	(2.5)
e)	What is hyperparameter tuning? Give one example of a hyperparameter in deep learning.	(2.5)
f)	*Why are baseline methods and performance metrics important while training and evaluating deep learning models?*	(2.5)
g)	What is transfer learning? Why is it useful in practice?	(2.5)
h)	What is meant by explainability in deep learning? Why is it important in sensitive applications?	(2.5)
	UNIT I	
Q2	Explain the following concepts with suitable examples and discuss their role in training deep neural networks: a) regularization, b) bias-variance trade-off, c) weight initialization strategies, and d) early stopping.	(10)
Q3	(a) Explain the idea of learning vs optimization in deep learning. Why should these two not be treated as the same? (b) Discuss Gradient Descent with Momentum. How does it differ from ordinary Gradient Descent, and why can it lead to faster convergence?	(4) (6)

Scanned with OKEN Scanner

	UNIT II	
Q4	Discuss the working principle, strengths, and limitations of any two of the following optimization methods used for training neural networks: Adagrad, Adadelta, RMSProp, Adam, NAG.	(10)
Q5	(a) What is the saddle point problem in neural networks? Why does it make optimization difficult?	(4)
	(b) Compare dropout, drop connect, and batch normalization as methods for improving training and generalization in deep learning.	(6)
	UNIT III	
Q6	A binary classifier for disease detection produces the following results on a test set: **TP = 80, TN = 150, FP = 20, FN = 50** Calculate the following and interpret them briefly: **Accuracy, Precision, Recall, and F1-score**.	(10)
Q7	(a) What are practical tips for training deep neural networks? Mention any four such tips.	(4)
	(b) Explain hyperparameter tuning with reference to manual vs automatic tuning and grid search vs random search. Which approach is generally more suitable in high-dimensional settings, and why?	(6)
	UNIT IV	
Q8	Explain the following with suitable examples: Autoencoders (undercomplete vs regularized), representation learning, and greedy pretraining.	(10)
Q9	(a) What is a Generative Adversarial Network (GAN)? Name its two major components and state their roles.	(4)
	(b) Discuss the importance of transfer learning, explainability, and ethics in deep learning-based systems deployed in real-world applications.	(6)

Scanned with OKEN Scanner